

CS 534 Individual Project Assignment 3: Summary

Explainable Artificial Intelligence for Predictive Maintenance Applications

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1. Purpose and Motivation

Predictive-maintenance (PdM) systems detect incipient machine failures but often rely on “black-box” models that operators distrust. This paper:

1. Provides a **public, realistic synthetic PdM dataset** (10 000 records).
2. Compares two **explainability strategies**—shallow decision-tree surrogates vs. a model-agnostic feature-deviation interface—to turn predictions into human-understandable explanations. :contentReference[oaicite:1]{index=1}

2. Data Set Construction

- **Size & Format:** 10 000 rows × (6 features + 1 label).
- **Features:**
 - `productID` (quality tier L/M/H)
 - Air temperature (K)
 - Process temperature (K)
 - Rotational speed (rpm)
 - Torque (Nm)
 - Tool-wear (min)
- **Failure Modes (independent rules):**
 1. **TWF** – tool-wear ≥ 200 –240 min
 2. **HDF** – $\Delta T < 8.6$ K **and** speed < 1380 rpm

3. **PWF** – power (torque × speed) outside [3500, 9000] W
 4. **OSF** – torque × wear > tier-specific limit (11 000–13 000 min·Nm)
 5. **RNF** – 0.1 % random failures
- **Imbalance:** 339 failures (3.39 %) :contentReference[oaicite:2]{index=2}

3. Baseline (“Black-Box”) Classifier

- **Model:** Bagged-trees ensemble with a 30× higher cost on false negatives.
- **Performance (5-fold CV):**
 - **Recall (failures):** 86.7 %
 - **Specificity (normal):** 98.7 %
- **Feature Importance:** Torque & rotational speed dominate; temperatures and `productID` contribute little. :contentReference[oaicite:3]{index=3}

4. Explanation Approaches

Approach	Mechanism	Pros	Cons
Shallow Decision Trees	15 trees (≤ 4 nodes) trained on feature-subsets; for each prediction, use the tree with highest global importance	• Clear, rule-based thresholds • Intuitive	• No explanation if no tree matches (4/20 cases)
Feature-Deviation Interface	Z-score each feature for the queried sample; report the two largest absolute deviations	• Always yields an explanation • Model-agnostic	• Often only “partially useful” (flags one irrelevant)

Evaluation on 20 stratified points:

- **Decision Trees:** 9 very useful; 2 partially; 4 none; 5 not-applicable (misclassified)
- **Feature Deviations:** 6 very useful; 14 partially useful; 0 none :contentReference[oaicite:4]{index=4}

5. Key Findings

1. **Quality vs. Coverage Trade-off:**

- Decision trees provide richer, rule-based explanations but may fail to mimic the ensemble (no explanation).
- Feature deviations always explain something, offering safe fallback at lower precision.

2. **Data-Design Impact:** Failure modes involving rarely used features (e.g. `productID`) are harder to classify and explain.

3. **Practical Recommendation:** Show decision-tree explanation when available; otherwise default to feature deviations. `:contentReference[oaicite:5]{index=5}`

6. Limitations & Future Work

- Only two XAI methods studied; plan to evaluate **LIME**, **SHAP**, etc.
 - Investigate dynamic thresholding in decision trees to reduce “no explanation” cases.
 - Validate findings on **real-world PdM logs** beyond synthetic data. `:contentReference[oaicite:6]{index=6}`
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7. Take-Away for CS 534

- Understand **surrogate models** vs. **feature-attribution** explainers.
- Use **human-centric metrics** (very/partial/limited/none) to evaluate explanations.
- **Combine methods** in deployment to balance interpretability quality with robustness. `:contentReference[oaicite:7]{index=7}`